

James Cranley

BM BCh MA MRCP PhD

Cambridge Stem Cell Institute, Jeffrey Cheah Biomedical Centre, Puddicombe Way, UK, CB2 0AW

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Summary

I hold an NIHR-funded clinical lecturer post at the University of Cambridge. This affords me 50% time for post-doctoral research (Cambridge Stem Cell Institute) and 50% time to complete Cardiology training (Royal Papworth Hospital). I am due to complete cardiology training in October 2027, sub-specialising in electrophysiology and inherited conditions. My ambition is to be a clinician-scientist leading a world-class research programme aimed at identifying novel therapeutic targets for cardiac arrhythmias.

Education

University of Cambridge

Cambridge, UK

PHD IN GENOMICS

2020 - 2025

- Funder: Wellcome Trust
- Advisor: Prof. Sarah Teichmann
- Thesis: *Spatially-resolved multiomic profiling of the human heart in adulthood and development*

University of Oxford

Oxford, UK

MEDICAL SCHOOL

2007 - 2013

- BM BCh (Distinction)
- BSc (1st class, joint top in year)
- Prizes: Old Member's Scholarship, Matilda Tambyraja (Obstetrics)

Eton College

Windsor, UK

SECONDARY SCHOOL

2002 - 2007

- A Levels: Maths, Further Maths, Biology, Chemistry, French, Latin, Greek

Clinical Experience

- 2025- **NIHR Academic Clinical Lectureship in Cardiology**, University of Cambridge
- 2015-2020 **NIHR Academic Clinical Fellowship in Cardiology**, East of England
- 2013-2015 **Foundation Training**, North-West Thames Deanery

GMC 7411765 | NTN EAN/007/00170/A

Research Experience

Post-doctoral (NIHR ACL)

Stem Cell Institute, University of Cambridge

Cambridge, UK

ADVISOR: PROF. SARAH TEICHMANN

2025-present

- Project: Linking single-cell resolved function and genomics.

Doctorate

Wellcome Sanger Institute

Hinxton, UK

ADVISOR: PROF. SARAH TEICHMANN

2020-2025

- Project: I used single-cell and spatial genomics to create cell atlases of the adult and foetal human heart.

Pre-doctoral (NIHR ACF)

Queen Mary University of London

London, UK

ADVISOR: PROF. PATSY MUNROE

2017

- Project: GWAS into modulators of the PR interval (ECG parameter).

Harvard University

Boston, MA, USA

ADVISOR: PROF. CALUM MACRAE

2016

- Project: Investigating non-canonical indices of coronary disease risk.

University of Cambridge

Cambridge, UK

ADVISORS: PROF CHRIS HUANG

2015

- Project: Patch clamping to investigate the effect of *Pgc1 β* deficiency on the cardiac sodium current.

Undergraduate (intercalated BSc)

University of Oxford

Oxford, UK

SUPERVISORS: PROF. NEIL HERRING, PROF. DAVID PATERSON

2011

- Project: The role of Galanin in cardiac autonomic control.

Publications

H-index 13 | Total citations 1153 | ORCID [0000-0002-0408-5801](https://orcid.org/0000-0002-0408-5801) | [Google Scholar](#) | [PhD Thesis](#)

IN REVIEW

Cranley J, Kanemaru K, Bayraktar S, Knight-Schrijver V, Hulbert R, Lana-Elola E, Aoidi R, Pett JP, Wilbrey-Clark A, Polanski K, Dabrowska M, Mulas I, Johnson H, Richardson L, Semprich CI, Kapuge R, Perera S, He X, Ho SY, Yayon N, Tuck L, Roberts K, Palmer JA, Davaapil H, Gambardella L, Patel M, Tyser R, Bernardo AS, Tybulewicz VLJ, Sinha S, Teichmann SA. 2024. Developmental Dynamics of Human Cardiogenesis: A multi-omic reference and its disruption in Trisomy 21. <https://doi.org/10.1101/2024.04.29.591736>. Under review at *Nature*.

Bayraktar S, **Cranley J**, Kanemaru K, Knight-Schrijver V, Colzani M, Davaapil H, Lee JCM, Polanski K, Richardson L, Semprich CI, Kapuge R, Dabrowska M, Mulas I, Perera S, Patel M, Ho SY, He X, Tyser R, Gambardella L, Teichmann SA, Sinha S. 2024. High-resolution atlas of the developing human heart and the great vessels. <https://doi.org/10.1101/2024.04.27.591127>. Under review at *Nature Cardiovascular Research*.

Pett JP, Prete M, Pham D, England N, Yuan H, Prigmore E, Tuck L, Oszlanczi A, To K, Xu C, Suo C, Dann E, He P, Kedlian V, Kanemaru K, **Cranley J**, Yang L, Elmentaite R, Oliver AJ, Cujba AM, Cakir B, Murray S, Mahbubani KT, Saeb-Parsy K, Gambardella L, Kasper M, Haniffa M, Nawijn MC, Teichmann SA, Meyer KB. 2025. An atlas of TF driven gene programs across human cells. <https://doi.org/10.1101/2025.05.30.657075>

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Nicastro M, Vermeer AM, Postema PG, Tadros R, Bowling FZ, Aegisdottir HM, Tragante V, Mach L, Postma AV, Lodder EM, van Duijvenboden K, Zwart R, Beekman L, Wu L, Jurgens SJ, van der Zwaag PA, Alders M, Allouba M, Aguib Y, Santome JL, de Una D, Monserrat L, Miranda AM, Kanemaru K, **Cranley J**, van Zeggeren IE, Aronica EM, Ripolone M, Zanotti S, Sveinbjornsson G, Ivarsdottir EV, Hólm H, Guðbjartsson DF, Skúladóttir ÁT, Stefánsson K, Nadauld L, Knowlton KU, Ostrowski SR, Sørensen E, Vesterager Pedersen OB, Ghouse J, Rand SA, Bundgaard H, Ullum H, Erikstrup C, Aagaard B, Bruun MT, Christiansen M, Jensen HK, Carere DA, Cummings CT, Fishler K, Tørring PM, Brusgaard K, Juul TM, Saaby L, Winkel BG, Mogensen J, Fortunato F, Comi GP, Ronchi D, van Tintelen JP, Nosedà M, Airola MV, Christiaans I, Wilde AA, Wilders R, Clur SA, Verkerk AO, Bezzina CR, Lahrouchi N. 2025. Bi-allelic variants in POPDC2 cause an autosomal recessive syndrome presenting with cardiac conduction defects and hypertrophic cardiomyopathy. *The American Journal of Human Genetics*, 112(7): 1681–1698. <https://doi.org/10.1016/j.ajhg.2025.04.016>

Troulé K, Petryszak R, Cakir B, **Cranley J**, Harasty A, Prete M, Tuong ZK, Teichmann SA, Garcia-Alonso L, Vento-Tormo R. 2025. CellPhoneDB v5: inferring cell–cell communication from single-cell multiomics data. *Nature Protocols*. <https://doi.org/10.1038/s41596-024-01137-1>

To K, Fei L, Pett JP, Roberts K, Blain R, Polański K, Li T, Yayon N, He P, Xu C, **Cranley J**, Moy M, Li R, Kanemaru K, Huang N, Megas S, Richardson L, Kapuge R, Perera S, Tuck E, Wilbrey-Clark A, Mulas I, Memi F, Cakir B, Predeus AV, Horsfall D, Murray S, Prete M, Mazin P, He X, Meyer KB, Haniffa M, Barker RA, Bayraktar O, Chédotal A, Buckley CD, Teichmann SA. 2024. A multi-omic atlas of human embryonic skeletal development. *Nature*, 635(8039): 657–667. <https://doi.org/10.1038/s41586-024-08189-z>

- Barnett SN, Cujba AM, Yang L, Maceiras AR, Li S, Kedlian VR, Pett JP, Polanski K, Miranda AMA, Xu C, **Cranley J**, Kanemaru K, Lee M, Mach L, Perera S, Tudor C, Joseph PD, Pritchard S, Toscano-Rivalta R, Tuong ZK, Bolt L, Petryszak R, Prete M, Cakir B, Huseynov A, Sarropoulos I, Chowdhury RA, Elmentaite R, Madissoon E, Oliver AJ, Campos L, Brazovskaja A, Gomes T, Treutlein B, Kim CN, Nowakowski TJ, Meyer KB, Randi AM, Nosedo M, Teichmann SA. 2024. An organotypic atlas of human vascular cells. *Nature Medicine*, 30(12): 3468–3481. <https://doi.org/10.1038/s41591-024-03376-x>
- Brunet J, Cook AC, Walsh CL, **Cranley J**, Tafforeau P, Engel K, Arthurs O, Berruyer C, Burke O’Leary E, Bellier A, Torii R, Werlein C, Jonigk DD, Ackermann M, Dollman K, Lee PD. 2024. Multidimensional Analysis of the Adult Human Heart in Health and Disease Using Hierarchical Phase-Contrast Tomography. *Radiology*, 312(1). <https://doi.org/10.1148/radiol.232731>
- Kanemaru K, **Cranley J**, Muraro D, Miranda AMA, Ho SY, Wilbrey-Clark A, Patrick Pett J, Polanski K, Richardson L, Litvinukova M, Kumasaka N, Qin Y, Jablonska Z, Semprich CI, Mach L, Dabrowska M, Richoz N, Bolt L, Mamanova L, Kapuge R, Barnett SN, Perera S, Talavera-López C, Mulas I, Mahbubani KT, Tuck L, Wang L, Huang MM, Prete M, Pritchard S, Dark J, Saeb-Parsy K, Patel M, Clatworthy MR, Hübner N, Chowdhury RA, Nosedo M, Teichmann SA. 2023. Spatially resolved multi-omics of human cardiac niches. *Nature*, 619(7971): 801–810. <https://doi.org/10.1038/s41586-023-06311-1>
- Knight-Schrijver VR, Davaapil H, Bayraktar S, Ross ADB, Kanemaru K, **Cranley J**, Dabrowska M, Patel M, Polanski K, He X, Vallier L, Teichmann S, Gambardella L, Sinha S. 2022. A single-cell comparison of adult and fetal human epicardium defines the age-associated changes in epicardial activity. *Nature Cardiovascular Research*. <https://doi.org/10.1038/s44161-022-00183-w>
- Ding WY, **Cranley J**, Begley D, Rao A, Snowdon RL, Mellor G, Gupta D. 2022. Impact of COVID-19 pandemic on cardiac rhythm management services: Views from the United Kingdom. *Heart Rhythm O2*, 3(5): 536–541. <https://doi.org/10.1016/j.hroo.2022.05.013>
- Reichart D, Lindberg EL, Maatz H, Miranda AMA, Viveiros A, Shvetsov N, Gärtner A, Nadelmann ER, Lee M, Kanemaru K, Ruiz-Orera J, Strohmer V, DeLaughter DM, Patone G, Zhang H, Woehler A, Lippert C, Kim Y, Adami E, Gorham JM, Barnett SN, Brown K, Buchan RJ, Chowdhury RA, Constantinou C, **Cranley J**, Felkin LE, Fox H, Ghauri A, Gummert J, Kanda M, Li R, Mach L, McDonough B, Samari S, Shahriaran F, Yapp C, Stanasiuk C, Theotokis PI, Theis FJ, Bogaerdt AVD, Wakimoto H, Ware JS, Worth CL, Barton PJR, Lee YA, Teichmann SA, Milting H, Nosedo M, Oudit GY, Heinig M, Seidman JG, Hubner N, Seidman CE. 2022. Pathogenic variants damage cell composition and single cell transcription in cardiomyopathies. *Science*, 377(6606). <https://doi.org/10.1126/science.abo1984>
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- Li J, Christodoulidou A, **Cranley J**, Ara F, Costopoulos C, Costanzo P, O’Sullivan M, Davies W, Densem C, Martin CA. 2021. Identifying predictive risk factors for permanent pacemaker implantation up to 30 days post-TAVI. *European Heart Journal*, 42(Supplement_1). <https://doi.org/10.1093/eurheartj/ehab724.2181>
- Ntalla I, Weng LC, Cartwright JH, Hall AW, Sveinbjornsson G, Tucker NR, Choi SH, Chaffin MD, Roselli C, Barnes MR, Mifsud B, Warren HR, Hayward C, Marten J, **Cranley J**, Concas MP, Gasparini P, Boutin T, Kolcic I, Polasek O, Rudan I, Araujo NM, Lima-Costa MF, Ribeiro ALP, Souza RP, Tarazona-Santos E, Giedraitis V, Ingelsson E, Mahajan A, Morris AP, Del Greco M F, Foco L, Gögele M, Hicks AA, Cook JP, Lind L, Lindgren CM, Sundström J, Nelson CP, Riaz MB, Samani NJ, Sinagra G, Ulivi S, Kähönen M, Mishra PP, Mononen N, Nikus K, Caulfield MJ, Dominiczak A, Padmanabhan S, Montasser ME, O’Connell JR, Ryan K, Shuldiner AR, Aeschbacher S, Conen D, Risch L, Thériault S, Hutri-Kähönen N, Lehtimäki T, Lyytikäinen LP, Raitakari OT, Barnes CLK, Campbell H, Joshi PK, Wilson JF, Isaacs A, Kors JA, van Duijn CM, Huang PL, Gudnason V, Harris TB, Launer LJ, Smith AV, Bottinger EP, Loos RJF, Nadkarni GN, Preuss MH, Correa A, Mei H, Wilson J, Meitinger T, Müller-Nurasyid M, Peters A, Waldenberger M, Mangino M, Spector TD, Rienstra M, van de Vegte YJ, van der Harst P, Verweij N, Kääb S, Schramm K, Sinner MF, Strauch K, Cutler MJ, Fatkin D, London B, Olesen M, Roden DM, Benjamin Shoemaker M, Gustav Smith J, Biggs ML, Bis JC, Brody JA, Psaty BM, Rice K, Sotoodehnia N, De Grandi A, Fuchsberger C, Pattaro C, Pramstaller PP, Ford I, Wouter Jukema J, Macfarlane PW, Trompet S, Dörr M, Felix SB, Völker U, Weiss S, Havulinna AS, Jula A, Sääksjärvi K, Salomaa V, Guo X, Heckbert SR, Lin HJ, Rotter JI, Taylor KD, Yao J, de Mutsert R, Maan AC, Mook-Kanamori DO, Noordam R, Cucca F, Ding J, Lakatta EG, Qian Y, Tarasov KV, Levy D, Lin H, Newton-Cheh CH, Lunetta KL, Murray AD, Porteous DJ, Smith BH, Stricker BH, Uitterlinden A, van den Berg ME, Haessler J, Jackson RD, Kooperberg C, Peters U, Reiner AP, Whitsel EA, Alonso A, Arking DE, Boerwinkle E, Ehret GB, Soliman EZ, Avery CL, Gogarten SM, Kerr KF, Laurie CC, Seyerle AA, Stilp A, Assa S, Abdullah Said M, Yldau van der Ende M, Lambiase PD, Orini M, Ramirez J, Van Duijvenboden S, Arnar DO, Gudbjartsson DF, Holm H, Sulem P, Thorleifsson G, Thorolfsson RB, Thorsteinsdottir U, Benjamin EJ, Tinker A, Stefansson K, Ellinor PT, Jamshidi Y, Lubitz SA, Munroe PB. 2020. Multi-ancestry GWAS of the electrocardiographic PR interval identifies 202 loci underlying cardiac conduction. *Nature Communications*, 11(1). <https://doi.org/10.1038/s41467-020-15706-x>
- Cranley J**, Hardiman A, Freeman LJ. 2020. Pulsed Levosimendan in advanced heart failure due to congenital heart disease: a case series. *European Heart Journal - Case Reports*, 4(3): 1–6. <https://doi.org/10.1093/ehjcr/ytaa080>

- Cranley J**, Krishnan U, Tweed K, Duehmke RM. 2019. Catastrophic antiphospholipid syndrome causing ST-segment elevation myocardial infarction with non-obstructive coronary arteries. *BMJ Case Reports*, 12(1): bcr-2018-225495. <https://doi.org/10.1136/bcr-2018-225495>
- Floré V, Brown AJ, Giblett JP, Liou K, **Cranley J**, Hoole SP, West NE. 2019. Clinical outcomes of bioresorbable vascular scaffolds implanted with routine versus selective optical coherence tomography guidance: results from a single-centre experience. *EuroIntervention*, 14(17): 1776–1783. <https://doi.org/10.4244/eij-d-18-00262>
- Cranley J**, MacRae CA. 2018. A New Approach to an Old Problem: One Brave Idea. *Circulation Research*, 122(9): 1172–1175. <https://doi.org/10.1161/circresaha.118.310941>
- Cranley J**, Bhagra C. 2018. Regulatory and occupational considerations in cardiology. *Medicine*, 46(11): 716–721. <https://doi.org/10.1016/j.mpmed.2018.08.003>
- Ahmad S, Valli H, Chadda KR, **Cranley J**, Jeevaratnam K, Huang CLH. 2018. Ventricular pro-arrhythmic phenotype, arrhythmic substrate, ageing and mitochondrial dysfunction in peroxisome proliferator activated receptor- γ coactivator-1 β deficient (Pgc-1 β) murine hearts. *Mechanisms of Ageing and Development*, 173: 92–103. <https://doi.org/10.1016/j.mad.2018.05.004>
- Brown AJ, Krishnan U, Giblett J, **Cranley JJ**, Bennett M, West N, Hoole S. 2016. HEMODYNAMIC LESION SIGNIFICANCE IS ASSOCIATED WITH HIGH PLAQUE BURDEN BUT NOT WITH VULNERABLE PLAQUE COMPOSITION. *Journal of the American College of Cardiology*, 67(13): 373. [https://doi.org/10.1016/s0735-1097\(16\)30374-6](https://doi.org/10.1016/s0735-1097(16)30374-6)
- Floré V, Brown A, **Cranley J**, Giblett J, Hoole S, West N. 2016. TCT-405 Optical coherence tomography-guided optimal bioresorbable vascular scaffold implantation informs subsequent implantation without intravascular imaging: further evidence for a 'learning curve'?. *Journal of the American College of Cardiology*, 68(18): B164–B165. <https://doi.org/10.1016/j.jacc.2016.09.540>
- Gill D, Bruce DJ, Ponsford MJ, **Cranley J**, Goodacre TE. 2013. Long-term follow-up of patients undergoing free tissue transfer to the lower limb following trauma. *European Journal of Plastic Surgery*, 36(7): 431–442. <https://doi.org/10.1007/s00238-013-0826-3>
- Herring N, **Cranley J**, Lokale MN, Li D, Shanks J, Alston EN, Girard BM, Carter E, Parsons RL, Habecker BA, Paterson DJ. 2012. The cardiac sympathetic co-transmitter galanin reduces acetylcholine release and vagal bradycardia: Implications for neural control of cardiac excitability. *Journal of Molecular and Cellular Cardiology*, 52(3): 667–676. <https://doi.org/10.1016/j.yjmcc.2011.11.016>
- Herring N, **Cranley J**, Lokale MN, Habecker B, Paterson DJ. 2010. Galanin reduces cardiac vagal acetylcholine release and bradycardia via a GalR1, protein kinase C dependent pathway. *The FASEB Journal*, 24(S1). https://doi.org/10.1096/fasebj.24.1_supplement.625.11

Qualifications

- 2025 **Clinical Research, Education & Leadership module**, University of Cambridge
- 2019 **Post-graduate Certificate in Medical Education (PGCME)**, University of Cambridge
- 2018 **Adult Life Support (ALS) Instructor**, Resuscitation Council (UK)
- 2015 **Membership (MRCP)**, Royal College of Physicians (London)

Presentations

- Risk stratification in Brugada Syndrome: a case presentation.* **British Cardiac Society**. 2025. Manchester, UK.
- The Devil is in the Detail: nodofascicular VT.* **British Heart Rhythm Society**. 2025. Edinburgh, UK.
- What you're missing matters: integrating short and long-read scRNA-seq.* **Oxford Nanopore Symposium**. 2024. Oxford, UK.
- Spatial Transcriptomics in Human Cell Atlasing.* **SpatialBiology 2023**. 2023. Online.
- Gene regulatory networks underlying cardiogenesis.* **European Society of Cardiology Congress**. 2023. Amsterdam, NL.
- A spatially-resolved multiomic atlas of human cardiac development.* **HCA General Meeting**. 2023. Toronto, CA.
- The adult and developing heart, one cell at a time.* **The Royal Society**. 2023. London, UK.
- The Hopes and Fears Lab (Public Engagement event).* **Cambridge Alumni Festival**. 2022. Cambridge, UK.
- Single-cell atlasing of the heart.* **BHF Symposium**. 2022. Online.

A novel variant causing JLNS, but not LQT1. **Heart Rhythm Congress.** 2019. Birmingham, UK.

Effusive-constrictive pericarditis – a complication of PCI. **British Society of Cardiac Imaging.** 2019. Cambridge, UK.

Haematoma risk post-angiography. **Society for Cardiac Interventions & Angiography (SCAI).** 2014. San Diego, USA.

Teaching Experience

Adult Life Support (ALS) teaching. **Resuscitation Council UK.** 2025. Nationwide, UK.

Core Cardiology course for general medics (Course Director). **Royal Papworth Hospital.** 2025. Cambridge, UK.

Cardiology Webinars for trainees (BJCA.tv series). **East of England Deanery.** 2020. Online.

Undergraduate Physiology supervisions (HOM). **Gonville & Caius College, University of Cambridge.** 2017. Cambridge, UK.

Medical Student Teaching Programme (teaching schedule for rotating students). **Harefield Hospital.** 2015. London, UK.

Management, Leadership & Responsibilities

Medical Director's Office Fellowship. This 12 month post gave me the ability to attend clinical governance meetings at Royal Papworth Hospital, giving me insight into the interface between clinicians and hospital management. As part of this I completed the Clinical Research, Education & Leadership (CREL) module at the Institute of Continuing Education as well as completing a Quality Improvement Project into improving and equalising registrar training opportunities.

Cardiology rota co-ordinator. In this 12 month post I was responsible for constructing the termly and weekly rotas and liaison with the consultant body. During this role I gained experience of personnel management, navigated 2 strikes, filled a large rota gap, and successfully proposed a business case and created a job advert for a clinical fellow.

BJCA representative to the BHRS. I represented the BJCA interest (electrophysiology and devices trainees) at BHRS committee meetings. I also attended BJCA committee meetings.

Trainee representative for East of England (Digital & Training Days). I constructed a website for cardiology trainees as well as organising regional training events.